

1. Which one is used for syntax analysis?
 - a) Parser
 - b) Scanner
 - c) Linker
 - d) Loader

The correct Answer is: a. Parser

2. Appropriate data structure for prioritizing seriously sick patients:
 - a) Stack
 - b) Tree
 - c) Priority Queue
 - d) Queue

The correct Answer is: c. Priority Queue

3. Different computer network role:
 - a) Routers
 - b) Network printer
 - c) Wireless access point
 - d) Hubs

The correct Answer is: b. Network printer

4. Best description of a database management system (DBMS):
 - a) Assisting image processing
 - b) Creating and maintaining a database
 - c) Helping manage computer's activity
 - d) Assisting document processing

The correct Answer is: b. Creating and maintaining a database

5. Attackers motivated by financial gain and performing sabotage, IP theft, and secrets theft:
 - a) Competitor
 - b) Hactivist
 - c) Activist
 - d) Stealer

The correct Answer is: a. Competitor

6. Output of JavaScript statement `Math.sqrt(36)`:
 - a) 6
 - b) 1
 - c) 30
 - d) 36

The correct Answer is: a. 6

7. False statement about programming language generations:

- a) Fifth-generation programming languages require powerful hardware and software.
- b) Assembly language should be translated to machine language before execution.
- c) Assembly language is machine-independent.
- d) Machine language can be executed without translation.

The correct Answer is: c. Assembly language is machine-independent.

8. Asymptotic notation for average case analysis:

- a) $f(n) = \Theta(\text{Theta})$
- b) $f(n) = \text{Big-Oh}$
- c) $f(n) = \omega$
- d) $f(n) = \Omega$

The correct Answer is: a. $f(n) = \Theta(\text{Theta})$

9. Access modifier to allow subclasses but not non-subclasses to access data fields/methods in the super-class:

- a) Default
- b) Private
- c) Public
- d) Protected

The correct Answer is: d. Protected

10. Correct way of creating a String object:

- a) `String m = new String("Technology");`
- b) `String m = 'H', 'e', 'l', 'l', 'o';`
- c) `String m = new String("Exit Exam");`
- d) `String m = "Computer Science!"`

The correct Answer is: `String m = "Computer Science!"`

11. Register holding the address of the next pair of instructions to be fetched from memory:

- a) Program counter
- b) Memory address register
- c) Memory buffer register
- d) Instruction buffer register

The correct Answer is: a. Program counter

12. Self-driving car acting in an environment:

- a) Stochastic
- b) Fully observable
- c) Discrete
- d) Partially observable

The correct Answer is: d. Partially observable

13. Software testing phase for demonstrating system satisfaction of client requirements:

- a) Acceptance testing
- b) Unit testing
- c) Integration testing
- d) System testing

The correct Answer is: a. Acceptance testing

14. Problem occurring when one transaction updates a database item but fails, and another transaction accesses the incorrect value:

- a) Temporary read problem
- b) Incorrect summary problem
- c) Dirty read
- d) Lost update problem

The correct Answer is: c. Dirty read

15. Assessment of the number of completed processes per hour:

- a) Turnaround time
- b) System throughput
- c) Waiting time
- d) Response time

The correct Answer is: b. System throughput

16. ----- is the whole that an agent has Observed so concerning the existing scenario in the environment:

- a) Performance measure
- b) Action
- c) Knowledge
- d) Perception

The correct Answer is: d. Perception

17. Program data independence in the database approach:

- a) Data and applications defined separately
- b) Programs can create redundant data independently
- c) The application may be affected when data changes
- d) Program and data defined together

The correct Answer is: a. Data and applications defined separately

18. Strategy for major changes to the architecture of a software system:

- a) Software maintenance
- b) Software reengineering
- c) Preventative maintenance
- d) Architectural transformation

The correct Answer is: b. Software reengineering

19. Data structure used for Depth First Traversal algorithm:

- a) Priority queue
- b) Tree
- c) Stack
- d) Queue

The correct Answer is: c. Stack

20. What is the output of the following C++ fragment code?

```
int a, b;
```

```
a= 13;
```

```
b = 9;
```

```
while((20 && 0) && (a> b))
```

```
cout<<"Plants are our life";
```

- a) No message will be displayed
- b) It displays message "Plants are our life" 20 times
- c) It displays message "Plants are Our Life" 4 times
- d) It displays message Plants are our life" infinitely

The correct Answer is: a. No message will be displayed

21. Phase done before semantic analysis in the compiling process:

- a) Code optimization
- b) Code generation
- c) Syntax analysis
- d) Intermediate code generation

The correct Answer is: c. Syntax analysis

22. Final phase in the compiling process:

- a) Code optimization
- b) Semantic analysis
- c) Intermediate code generation
- d) Code generation

The correct Answer is: d. Code generation

23. Which one is False about Relationship between computer architecture and organization:

- a) An organization can last as long as an architecture does
- b) An architecture can remain for a long time, but its organization can change
- c) An architecture can be used for a long time, encompassing different computer models
- d) Manufacturers offer computers with the same architecture but different organization

The correct Answer is: a. An organization can last as long as an architecture does

24. Which one is the Hardest environment for an agent:

- a) Sequential, non-deterministic, dynamic
- b) Partially observable, static, continuous
- c) Fully observable, sequential, deterministic
- d) Deterministic, fully observable, static

The correct Answer is: a. Sequential, non-deterministic, dynamic

25. Separating some machines from the rest to shield them from possible attacks:

- a) Switch
- b) Router
- c) Network operating system
- d) Firewall

The correct Answer is: d. Firewall

26. Extending specialized classes from generalized classes in object-oriented programming:

- a) Inheritance
- b) Overloading
- c) Overriding
- d) Data field encapsulation

The correct Answer is: a. Inheritance

27. Network architecture characteristic allowing quick expansion without impacting existing users' services:

- a) Scalability
- b) Quality of service
- c) Security
- d) Fault tolerance

The correct Answer is: a. Scalability

28. Security service protecting information disclosure from unauthorized entities:

- a) Confidentiality
- b) Availability
- c) Integrity
- d) Authentication

The correct Answer is: a. Confidentiality

29. Step in problem solving that lists the actions and states the agent has to consider to achieve a goal:

- a) Goal formulation
- b) Execute
- c) Search
- d) Problem formulation

The correct Answer is: c. Search

30. Language generated from a given grammar $G: G = (\{S\}, \{a, b, c, d\}, S, P)$ Production: aSb/λ which one will not be generated by the grammar?

- a) a
- b) λ
- c) aabb
- d) ab

The correct Answer is: a. "a"

31. Type of data structure: Linked list

- a) Fixed-size data structure
- b) Static data structure
- c) Non-linear data structure
- d) Dynamic data structure

The correct Answer is: d. Dynamic data structure

32. Method used to obtain the meaning of encrypted information without the secret key:

- a) Decryption
- b) Encryption
- c) Cryptography
- d) Crypto-analysis

The correct Answer is: d. Crypto-analysis

33. Output of the following C++ code fragment:

```
for (int n = 1; n <= 18; n = n + 2)
{
    if (n % 7 != 0)
        cout << n;
    else
        break;
}
```

- a) 1 3 5 7 9 11 13 15 17
- b) 1 2 3 4 5 6
- c) 135
- d) 3 5 7

The correct Answer is: d. 3 5 7

34. Computer most secure compared to the rest:

- a) A computer connected to the Internet with strong intrusion detection
- b) A computer connected to the Internet with the latest anti-virus
- c) A computer not connected to the Internet
- d) A computer connected to the Internet with the latest firewall system

The correct Answer is: A computer not connected to the Internet

35. Network security issue indicating weakness in networks and devices:

- a) Availability
- b) Attack
- c) Vulnerability
- d) Threat

The correct Answer is: c. Vulnerability

36. Code optimization technique appropriate for modifications to ensure no new errors are introduced:

- a) Dead code elimination
- b) Arithmetic simplification
- c) Compile-time evaluation
- d) Code motion

The correct Answer is: a. Dead code elimination

37. Data structure responsible for controlling locks:

- a) Lock table
- b) Query optimizer
- c) Query processor
- d) Lock manager

The correct Answer is: d. Lock manager

38. Largest network type compared to the rest:

- a) The Internet
- b) Local Area Network
- c) Metropolitan Area Network
- d) Wide Area Network

The correct Answer is: d. Wide Area Network

39. Index of data value 19 in a given array:

`$list = array(20, 12, 23, 19, 18);`

- a) 2
- b) 3
- c) 1
- d) 0

The correct Answer is: b. 3

40. Meaning of Σ^* in a given input alphabet: Given input alphabet $\Sigma = \{a,b,c,d\}$ and λ

- a) $\Sigma^1 + U\Sigma^2 U\Sigma^3 U\Sigma^4$
- b) $\Sigma^+ U \{\lambda\}$
- c) λ
- d) Σ^+

The correct Answer is: b. $\Sigma^+ U \{\lambda\}$

41. Scope of a program where attributes are visible and can impact program execution:

- a) Arrangement
- b) Flow control
- c) Organization
- d) Architecture

The correct Answer is: d. Architecture

42. Asymptotic notation describing worst-case analysis:

- a) $f(n) = \Omega$
- b) $f(n) = o$
- c) $f(n) = \Theta$
- d) $f(n) = \omega$

The correct Answer is: b. $f(n) = o$

43. Testing level required to ensure modifications have not introduced new errors:

- a) Regression testing
- b) Dynamic testing
- c) Static testing
- d) Beta testing

The correct Answer is: Regression testing

44. Not a single-level ordered index:

- a) Primary index
- b) Secondary index
- c) Clustering index
- d) Multilevel index

The correct Answer is: d. Multilevel index

45. Problem occurring when two transactions accessing the same database item have their operations interleaved, resulting in an incorrect value:

- a) Temporary read problem
- b) Dirty read problem
- c) Incorrect summary problem
- d) Lost update problem

The correct Answer is: Lost update problem

46. Fan trap in an Entity Relationship Diagram:

- a) Highly likely to occur when more than two one-to-many relationships fan out from an entity
- b) Can be resolved by readjusting the model to eliminate one-to-many relationships fanning out from an entity
- c) Occurs when there is no pathway between related entities
- d) Occurs when there is an ambiguous pathway between some entity occurrences in related entity types

The correct Answer is: c. Occurs when there is no pathway between related entities

47. Costly and not recommended to use a database management system (DBMS)

- a) To deny access to unauthorized users
- b) In embedded systems with limited storage space
- c) To control redundancy
- d) In situations requiring multiple user access to the database

The correct Answer is: b. In embedded systems with limited storage space

48. False statement about POP3 and SMTP:

- a) Both are application layer protocols
- b) A sender can send emails using SMTP
- c) Both are network layer protocols
- d) A sender can receive emails using POP3

The correct Answer is: c. Both are network layer protocols

49. Transaction failure:

- a) Concurrency control enforcement avoids transaction failure
- b) Transactions fail if logical errors are detected
- c) Catastrophes can cause transaction failure
- d) Transactions fail if disk blocks lose their data

The correct Answer is: a. Concurrency control enforcement avoids transaction failure

50. Program used to decode scrambled text:

- a) Encryption program
- b) Ciphertext
- c) Decryption program
- d) Secret key

The correct Answer is: Decryption program

51. False statement about universal hashing technique:

- a) Universal hashing selects a hash function from a set of hash functions independent of keys
- b) Universal hashing selects a hash function randomly from a set of hash functions for each key
- c) Single fixed hash function technique is better than universal hashing technique in terms of collision reduction
- d) Universal hashing reduces the chance of data collision

The correct answer is: c. Single fixed hash function technique is better than universal hashing technique in terms of collision reduction

52. Smallest heading tag:

- a) <H3>
- b) <H1>
- c) <H6>
- d) <H4>

The correct answer is: <H6>

53. Data updates best describing the statement: "All modified data items in the cache are written either after a transaction ends its execution or after a predetermined number of transactions have concluded their execution":

- a) Deferred update
- b) In-place update
- c) Shadow update
- d) Immediate update

The correct answer is:

54. Asynchronous message communication:

- a) Blocking send
- b) Non-blocking receive
- c) Blocking receive
- d) Direct message

The correct answer is: b .Non-blocking receive

55. Output of the following JavaScript statement: Math.floor(5.9)

- a) 6
- b) 5
- c) 11.8
- d) 9

The correct answer is: b. 5

56. Time complexity order of the Quick sort algorithm:

- a) $O(n^2)$
- b) $O(n)$
- c) $O(2*n)$
- d) $O(n \log n)$

The correct answer is: d. $O(n \log n)$

57. Time complexity order of the binary searching algorithm:

- a) $O(n)$
- b) $O(1)$
- c) $O(\log n)$
- d) $O(n^3)$

The correct answer is: c. $O(\log n)$

58. Stage of software development process involving modification of software to reflect changing customer and market requirements:

- a) Evolution
- b) Validation
- c) Development
- d) Specification

The correct answer is: a. Evolution

59. Equivalent expression to Σ^0 ?

- a) Σ^+
- b) Σ
- c) $\{\lambda\}$
- d) Σ^*

The correct answer is: c.

60. Role of a human in system administration:

- a) Ethics is not a requirement to manage computers
- b) It doesn't require organizational skills
- c) System administrator is not expected to have skills and knowledge
- d) It requires patience, understanding, and knowledge

The correct answer is: d. It requires patience, understanding, and knowledge

61. Layer of the OSI model where devices such as bridges, switches, and Network Interface cards are used:

- a) Application layer
- b) Physical layer
- c) Network layer
- d) Data link layer

The correct answer is: d. Data link layer

62. Pop operation of a Stack data structure:

- a) Pop inserts a new element at the beginning/bottom of the stack
- b) Pop inserts a new element in the middle of the stack
- c) Pop removes the end/top element from the stack
- d) Pop returns the end/top element without deleting it

The correct answer is: c. Pop removes the end/top element from the stack

63. Type of variable accessible throughout the C++ program scope:

- a) Automatic
- b) Static
- c) Global
- d) Local

The correct answer is: c. Global

64. Term describing a university's network designed for access by academic staff, students, and administrative workers:

- a) Wide Area Network
- b) Local Area Network
- c) Metropolitan Area Network
- d) The Internet

The correct answer is: b. Local Area Network

65. Measure taken to ensure information and information systems' availability, integrity, authentication, and confidentiality:

- a) Interception
- b) Wiretapping
- c) Information security
- d) Information assurance

The correct answer is: c. Information security

66. Algorithm used to extract the Minimum Spanning Tree from a graph:

- a) Huffman encoding algorithm
- b) Dijkstra's algorithm
- c) Prim's algorithm
- d) Merge sort algorithm

The correct answer is: c. Prim's algorithm

67. False statement about Direct Address table and Hash Table data structures:

- a) Direct Address table doesn't use a hash function to map keys
- b) Hash table allocates one separate memory slot for each key in the universe U
- c) If the universe U is very large, a Hash table is better than a Direct Address table
- d) There is a one-to-one correspondence between keys in the universe U and memory slots in the Direct Address table

The correct answer is: d .There is a one-to-one correspondence between keys in the universe U and memory slots in the Direct Address table

68. Non-functional requirement of software systems:

- a) Memory requirement
- b) Displaying information
- c) Response time
- d) Reliability

The correct answer is: b. Displaying information

69. Branch of study dealing with whether a problem can be solved at all, regardless of the resources required:

- a) Complexity theory
- b) Computability theory
- c) Set theory
- d) Automata theory

The correct answer is: b. Computability theory

70. Type of Turing machine with two tapes, one read-only and the other read-write tape:

- a) Non-deterministic Turing machine
- b) Multi-head Turing machine
- c) Off-line Turing machine
- d) Multi-dimensional Turing machine

The correct answer is: b . Multi-head Turing machine

71. Deadlock prevention mechanisms not requiring a timestamp:

- a) Wait-die
- b) Wound-wait
- c) No-wait
- d) Wait-wait

The correct answer is: c. No-wait

72. Quantifier used for some portion of the universe:

- a) $\exists$
- b) $\forall$
- c) $\in$
- d) $\rightarrow$

The correct answer is: a. $\exists$

73. False statement about threads:

- a) It is a group of processes
- b) Threads are the entities scheduled for execution on the CPU
- c) Threads have program counters
- d) Threads have registers to hold their working memory

The correct answer is: a. It is a group of processes

74. Command used to check if a computer upstairs is connected to the network:

- a) nslookup
- b) DHCP discover
- c) traceroute
- d) ping

The correct answer is: d. ping

75. False statement about project planning stage:

- a) Cost estimation is done during project planning
- b) Preparing a time schedule is done during project planning
- c) Risk analysis is done during project planning
- d) Project planning is a one-time task in software development life cycle

The correct answer is: Project planning is a one-time task in software development life cycle

76. Size of large data files can be reduced to save storage disk space. Which algorithm is appropriate to reduce the size of large files?

- a) Prim's algorithm
- b) Huffman encoding algorithm
- c) Heap sort algorithm
- d) Merge sort algorithm

The correct answer is: b. Huffman encoding algorithm

77. What is the output of the following Java fragment code?

```
int[] list = new int[4];  
int sum = 0;  
  
for (int i = 0; i < list.length; i++) {  
    list[i] = 13;  
    sum += list[i];  
}  
  
System.out.print(sum);
```

- a) 3
- b) 6
- c) 18
- d) 36

The correct answer is: c. 18

78. Which one of the following is a two-level cache having an internal and external caches?

- a) Split cache
- b) Multilevel cache
- c) Unified cache
- d) Single level cache

The correct answer is: b. Multilevel cache

79. Which one of the following is true about functions in JavaScript?

- a) Function names can begin with digits
- b) Function names are not case-sensitive
- c) Function names can contain spaces
- d) There is no limit to the number of function parameters that a function may contain

The correct answer is: d. There is no limit to the number of function parameters that a function may contain

80. Which one of the following is not correct about packet and circuit switching?

- a) In packet switching, messages are sent in small blocks
- b) In packet switching, an end-to-end connection has to be established
- c) In circuit switching, a channel is dedicatedly used
- d) Packet switching is more efficient than circuit switching

The correct answer is: b. In packet switching, an end-to-end connection has to be established

81. An interface that has multiple lines connecting input/output modules and peripheral devices and assures multiple bits to be transferred at the same time is:

- a) Parallel interface
- b) Bus interface
- c) Serial interface
- d) One line interface

The correct answer is: a. Parallel interface

82. Among the phases of the compiling process, in which phase are a sequence of characters converted into a sequence of tokens?

- a) Code optimization
- b) Lexical analysis
- c) Semantic analysis
- d) Syntax analysis

The correct answer is: b. Lexical analysis

83. A process is terminated by the operating system due to a reference to non-existing memory. Which one of the following best explains the process termination?

- a) Error exit
- b) Normal exit
- c) Terminated by another process
- d) Fatal exit

The correct answer is: d. Fatal exit

84. In a tree data structure, if a node has no parent node, then the node is:

- a) Parent node
- b) Internal node
- c) External node
- d) Root node

The correct answer is: d. Root node

85. Which one of the following is not a delimiter of PHP code?

- a) `<script language="PHP">`
- b) `<?php`
- c) `?>`
- d) `<caption></caption>`

The correct answer is: d. `<caption></caption>`

86. Which one of the following is false about String in Java?

- a) We can create a String object using an array of characters
- b) We can create a String object using String literal
- c) The content of a string can be changed once the string is created
- d) In Java, a string is treated as an object

The correct answer is: The content of a string can be changed once the string is created

87. What is the output of the following Java fragment code?

```
int n = 6, m = 15, p = 3;
n -= m;
n *= k;
System.out.print(n);
```

- a) 3
- b) -27
- c) 27
- d) 24

The correct answer is: b. -27

88. Which one of the following is true about frames in HTML?

- a) Frames allow parts of the page to remain stationary while other parts scroll
- b) Load on the server is not affected if there are a large number of frames on a page
- c) All browsers support frames
- d) Frames are not difficult to handle for search engines

The correct answer is: a. Frames allow parts of the page to remain stationary while other parts scroll

89. Which one of the following is false about arrays in C++?

- a) We use `[]` square brackets at the time of array declaration
- b) The size of an array should be constant at the time of array declaration
- c) We can access elements of arrays without using an index number
- d) An array is a collection of similar data objects

The correct answer is: We can access elements of arrays without using an index number

90. Which one of the following is evaluating the degrees of success of an agent?

- a) Perception
- b) Knowledge
- c) Performance measure
- d) Action

The correct answer is: c. Performance measure

91. Which one of the following is not an example of scripting languages?

- a) Perl
- b) Python
- c) C++
- d) PHP

The correct answer is: c. C++

92. Which layer uses port numbers to identify applications?

- a) Physical layer
- b) Transport layer
- c) Application layer
- d) Network layer

The correct answer is: b. Transport layer

93. Given the schemas:

BOOK (Book_id, Title, Publisher_name)
BOOK_AUTHOR (Book_id, Author_name)
PUBLISHER (Name, Address, Phone)

Which one of the following SQL queries is correct to delete the publisher named 'XWZ' from the PUBLISHER table?

- a) DELETE FROM BOOK_AUTHOR WHERE Address = "XWZ"
- b) DELETE FROM PUBLISHER WHERE Name = 'XWZ'
- c) DELETE FROM PUBLISHER WHERE Address = "XWZ"
- d) DELETE FROM BOOK WHERE Name = 'XWZ'

The correct answer is: b. DELETE FROM PUBLISHER WHERE Name = 'XWZ'

94. Optimization problem on a graph:

- a) How many cycles are there in graph G?
- b) Find minimum spanning tree from graph G
- c) Traverse all vertices in graph G
- d) Is vertex p in V reachable from vertex q in V in graph G?

The correct answer is: Find minimum spanning tree from graph G

95. Incorrectly coupled generation and advancement in computer architecture:

- a) First generation -> single user
- b) Third generation -> batch system
- c) Second generation -> multiprogramming
- d) Fourth generation -> real-time systems

The correct answer is :

96. Valid statement regarding PHP variables:

- a. `$test = 9.5`
- b. `$test = "This is Computer science";`
- c. `$test = 5;`
- d. `$test = "Computer";`

The correct answer is :

97. Purpose of an opcode:

- a. Hold destination address of an operand
- b. Hold source address of an operand
- c. Determine the operation to be done
- d. Hold the next instruction reference

The correct answer is :

98. Comparison between Java and JavaScript:

- a. Both are object-oriented languages
- b. JavaScript can be used in sophisticated applications
- c. JavaScript is compiled and interpreted in source code form
- d. JavaScript has more features than Java

The correct answer is :